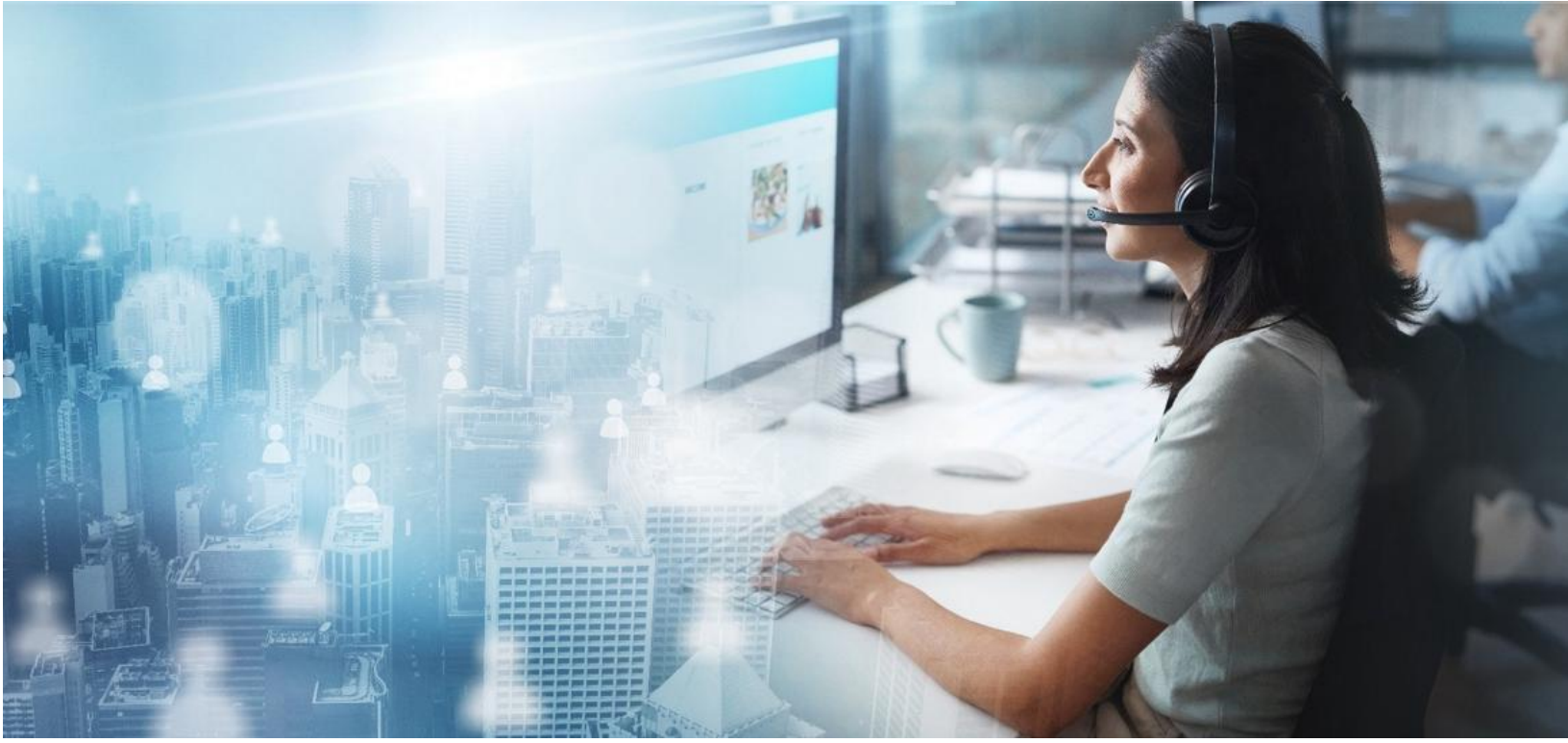

# Department of Health and Human Services (HHS), Centers for Medicare & Medicaid Services (CMS)

Agency/Docket Number: CMS-0042-NC
RIN: 0938-AV68

# Request for Information; Health Technology Ecosystem


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**Submitted to:**
https://www.regulations.gov
June 16, 2025


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June 16, 2025

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Submitted via www.regulations.gov

**RE: CMS-0042-NC, RIN: 0938-AV68; Health Technology Ecosystem RFI**

Dear Drs. Oz and Keane,

On behalf of Microsoft Corporation, we appreciate the opportunity to respond to the Request for Information (RFI) - Health Technology Ecosystem issued by the Centers for Medicare & Medicaid Services (CMS) and the Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC).

We commend CMS and ASTP/ONC for their leadership in advancing a patient-centered, digitally enabled healthcare system and welcome the chance to contribute to this important dialogue.

HHS, through CMS and ASTP/ONC, is uniquely positioned to ignite a system-wide evolution—one that redefines how care is delivered, experienced, and sustained across the nation. With the healthcare industry using digitized multimodal data, a population with ubiquitous access to smartphones, and the breakthrough capabilities of generative AI, HHS has an unprecedented opportunity to give individuals more influence and control over their own health, while improving outcomes and reducing burdens on providers.

By championing the responsible adoption of AI at scale, HHS can empower every individual to take charge of their health and every provider to supply the best care available. With forward-thinking policies, a robust data infrastructure, and a commitment to collaboration across the healthcare ecosystem, HHS can use new technologies to unlock innovation at every level. Together, these initiatives converge on a singular, powerful mission: to make healthcare safer and more accessible, efficient, and cost-effective for all. Microsoft is ready to assist the agencies with this important mission.



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By aligning financial incentives with innovation, HHS can send a powerful and unambiguous signal across the healthcare landscape: when an AI tool demonstrably improves care quality while reducing costs, its adoption will be recognized, supported, and rewarded. This will encourage technology developers to prioritize real-world clinical value, incentivize healthcare providers to integrate AI into their workflows, and reassure patients that these tools are designed to enhance and complement their care.

Microsoft welcomes the opportunity to support the federal government's efforts to improve the healthcare system through the responsible and innovative use of technology. Our response to this RFI reflects our cross-sector experience, our leadership in artificial intelligence and data science, and our longstanding engagement with healthcare stakeholders—including CMS, ASTP/ONC, providers, researchers, payers and patients.

Building on decades of research, product development, and collaboration with HHS and other public health and research leaders, Microsoft is pioneering next-generation technologies to help transform health and healthcare. We work at all levels of the stack, from collaborating with communities to establish and implement standardized APIs for health data access and workflows, to developing industry-leading health data platforms, healthcare-specific copilots, and developer solutions for healthcare-specific multimodal models and agentic systems. Our unified goal is to empower everyone in the healthcare ecosystem: each individual patient and their extended care team.

We believe that a connected health ecosystem should rest on **three foundational pillars**:

1. **Put the human at the center of care—** Empower patients to achieve their health goals and reduce the friction of navigating care and reimbursement, while taking care that implementation does not burden providers with unfunded mandates that take time and resources away from patient care.
2. **Promote evidence-based medicine—**Harness real-world data and insights, collected from everyday healthcare settings, to identify unwarranted care variation—care well-informed patients would not want—to drive the more effective and personalized treatment and prevention that is foundational to value-based care. Importantly, quality metrics should include measures of patient experiences, such as Patient-Reported Outcome Measures (PROMs), rewarding systems that measurably reduce friction and foster sustained, positive engagement.
3. **Drive choice and robust markets through industry standards and incentives—**Just as past policies successfully digitized medical data and advanced data interoperability, we now need architectural standards that enable an app-based marketplace that promotes strong patient choice mechanisms and powered by data portability and functional abstractions.

For each of these pillars, the emerging capabilities of generative and agentic AI will be indispensable. From automating API-based data access and structuring unstructured data for harmonization and analysis, to delivering high-level applications such as ambient clinical intelligence and automated second opinions for doctors and nurses or personalized care navigators for patients/consumers—AI has the potential to make medicine more precise,



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personal, affordable, and accessible. Our collective opportunity is to enable and connect this rapidly emerging marketplace in a way that empowers patients, enables clinicians to focus on medicine, not documentation, and anchors every decision on evidence.

But this vision requires a solid foundation of technical infrastructure, interoperability standards, and incentives. Certification reform must go beyond theoretical capabilities to validate data availability in deployed systems. We need trusted enforcement mechanisms that confirm facilities configure and maintain their APIs to expose comprehensive, usable data while protecting patient privacy. Performance requirements must confirm APIs work at scale, not only for individual record access but also for system-wide quality improvement and population health.

As transformative as it is, AI technology alone is not enough. Real progress will require robust infrastructure, trusted interfaces, and bold public-private partnerships that together meet the highest standards of accuracy, reliability, transparency, privacy, and compliance.

As detailed in the appendix in response to this RFI, we make the following recommendations centered on three main themes: 1) patient empowerment, 2) advancing foundational interoperability standards and infrastructure, and 3) promoting a competitive, learning ecosystem:

## Theme 1: Empower Patients with Comprehensive Data Access and Transparency in TEFCA

The greatest opportunity for health IT lies in **requiring API access to all Electronic Health Information (EHI)**, not only the United States Core Data for Interoperability (USCDI) elements, empowering patients with complete, usable access to their health information. This includes flowsheets, detailed nursing documentation, therapy notes, and vendor-specific data. We recommend mandating comprehensive export capabilities with appropriate documentation to enable patients and their authorized applications to access their complete health story, specifically:

- **Recommendation: *Require API Access to Full EHI.***
  **Why:** Patients require API access to all their EHI—including notes and images—in a computable format. This approach empowers patients and aligns with industry standards like the Argonaut EHI Export API for innovative applications.
- **Recommendation: *Require Programmatic Access to Medical Images.***
  **Why:** Diagnostic quality medical images should be accessible via standardized APIs using the same secure Substitutable Medical Applications and Reusable Technologies (SMART) on Fast Healthcare Interoperability Resources (FHIR) authorization as other health data. This allows images to be integrated into a patient's complete health record and used by innovative tools.

The Trusted Exchange Framework and Common Agreement (TEFCA) is designed to improve nationwide interoperability, but its current implementation treats patients as passive subjects of data exchange rather than active participants. While providers can query each other's records through TEFCA, patients have no visibility into who is accessing their information or why it is



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being shared, nor any ability to control these exchanges. **This opacity undermines trust and limits TEFCA's potential for patient empowerment**. To improve TEFCA:

- **Recommendation: *Empower Individuals with Transparency and Control in TEFCA.***
  **Why:** To build trust in national networks like TEFCA, individuals must have clear visibility (e.g., audit logs) and robust controls (e.g., opt-out, "ask me first," freeze access) over how their data is shared, managed at the network or QHIN level.
- **Recommendation: *Confirm a Trustworthy Architecture for TEFCA Individual Access.***
  **Why:** All services enabling individual data access through TEFCA (Individual Access Services) must operate under a high-assurance security model, requiring strong, federally recognized identity verification and explicit, verifiable patient consent before data sharing. This approach is fundamental for building trust and confirming patient control.
- **Recommendation: *Establish a TEFCA Patient-Developer Program.***
  **Why:** Many patients have technical skills to build their own health tools. A free, identity-verified credential that enables patients to develop applications accessing only their own data fosters patient-driven innovation and confirms individuals can participate directly in the health data ecosystem without intermediaries.

## Theme 2: Advance Foundational Interoperability Standards and Infrastructure

To achieve seamless and efficient data exchange across the healthcare ecosystem, it is essential to take the next major step in advancing foundational interoperability standards and infrastructure. This will enable better communication, coordination, and innovation in healthcare delivery, ultimately improving patient outcomes and reducing burdens on providers. More specifically:

- **Recommendation: *Establish Public Foundational Infrastructure for a National Provider Directory.***
  **Why:** A free, publicly available, and machine-readable national directory of healthcare provider/system electronic service endpoints and capabilities is needed. This simplifies connections and reduces friction for developers, providers, patients, and payers. Including provider performance metrics against regional benchmarks would further empower patients to make informed choices.
- **Recommendation: *Steward USCDI Development for Pragmatic Interoperability.***
  **Why:** An improved, evidence-based process for expanding the USCDI is essential. Clearer definitions and robust testing confirm this common data foundation is reliably implemented for better system communication.
- **Recommendation: *Keep Single-Patient API Certification Current.***
  **Why:** Certified Health IT must support current, secure versions of the SMART App Launch Framework. This allows patient applications to securely connect to EHRs and enables a wider range of innovative tools, including through full CORS support and offline access.


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- **Recommendation: *Confirm Current and Performant Bulk Data APIs.***
  **Why:** Standardized, efficient access to data for groups of patients via FHIR Bulk Data is crucial for population health, research, and analytics. This standard API must be kept current and perform as well as proprietary methods.
- **Recommendation: *Adopt FHIR Subscriptions for Event-Driven Workflows.***
  **Why:** FHIR Subscriptions enable applications to receive automatic notifications of health data changes (e.g., new lab results). This event-driven approach is more efficient and supports timely actions for patients and clinicians.

## Theme 3: Promote A Competitive, Learning Health Ecosystem

Using real-world data with advanced technologies enables CMS to model the impact of new policies and incentives across the entire health system, allowing CMS administrators to make informed decisions about which policies and incentives to implement. As the Dartmouth Atlas of Healthcare project showed, fostering innovation and confirming that health technology serves the diverse needs of patients and providers, especially in rural and underserved areas, requires insight into the financial flows of healthcare. This can be accomplished by integrating financial information into the available health data streams made available to patients and providers. Furthermore, by leveraging financial incentives, a broader array of both smaller and larger players in the health tech industry can be encouraged to develop solutions that enhance healthcare delivery. More specifically:

- **Recommendation: *Use Real-Time, Real-World Data to Assess Policy Impact.***
  **Why:** For decades, the Dartmouth Atlas project has shown that CMS data can reveal unwarranted care variation and up to 30% waste. The project provides a roadmap that consumers and policy makers can use to better deploy resources that support preference-sensitive decisions (e.g., well-informed patients seeking elective care), reduce variation from supply sensitive decisions (e.g. oversupply of hospital beds and diagnostic imaging centers), and reinforce effective care decisions (e.g., aspirin for heart attacks, mammograms, colonoscopies). Linking digital health data to claims and encounter data would enable CMS to assess downstream effects of treatments and outcomes over time with best-in-class outcomes and cost and would further support regional-based benchmarking of providers to inform patient choice.
- **Recommendation: *Enhance Blue Button 2.0 with CMS-Controlled Data Sources.***
  **Why:** Adding prior authorization information would help patients understand coverage decisions, track authorization status, and effectively appeal denials. CMS could also add a list of organizations and provider identifiers where beneficiaries have received care to help apps discover and connect to clinical data sources
- **Recommendation: *Offer Grants for FHIR Implementation.***
  **Why:** Financial support can be a powerful enabler of interoperability. CMS/CMMI can offer targeted grants or incentive programs to assist providers—especially small practices and rural health systems—in implementing FHIR standards.


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- **Recommendation: *Provide Hands-On Training and Skill Development.***
  **Why:** In addition to robust infrastructure, standards, and incentives, broad skill development is critical for sustained technology diffusion—but outcomes matter. HHS can create competitive programs that foster innovation in training methods and scale those that effectively empower staff to use technologies and tools that demonstrably improve patient outcomes.
- **Recommendation: *Co-Fund the Establishment of an Industry Association.***
  **Why:** We recommend HHS co-fund the launch of an industry association of health technology providers, with the aim of creating an effective organizing framework for industry-government partnerships that advance HHS's goals and evolve health tech infrastructure for all.

Microsoft brings a powerful and unique combination of secure, globally scaled infrastructure, deep healthcare domain expertise, longstanding leadership in the health standards and interoperability community, and a comprehensive suite of enabling technologies to help HHS realize its vision for a more efficient, effective, and intelligent health system. By aligning innovation with public policy goals, Microsoft is ready to support a connected ecosystem that delivers measurable value to patients, providers, and the nation.

We are committed to supporting this important journey and look forward to continued engagement.

Sincerely,

Peter Lee, PhD
President, Microsoft Research

Jim Weinstein, DO, MS
Vice President
Microsoft Research Health Futures

Josh Mandel, MD
Chief Architect for Health
Microsoft Research Health Futures

Jonathan Carlson, PhD
Vice President and Managing Director,
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# APPENDIX

***PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?*** *While basic structured data (USCDI) is increasingly available, much of the richest data remains difficult to access programmatically.*

***Readily Available & Valuable (Increasingly):***

- *USCDI data elements via FHIR APIs.*
- *Medicare claims data via Blue Button 2.0.*

***Valuable but Hard to Access (PC-8a):*** *Specific data types include:*

- *Diagnostic quality medical images.*
- *Full flowsheet data: Comprehensive view of patient status and interventions. Addressed by EHI export and potentially USCDI expansion.*
- *Detailed/granular lab results (e.g., cancer, microbiology): Beyond simple numerics, including narratives, structured reports, and interpretations.*
- *Schedules/appointment information: Programmatic access is rare.*
- *Patient-Reported Outcomes (PROs). Addressed by EHI export and USCDI expansion.*
- *Price information (patient-specific cost estimates).*

***PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?***

***a. What difficulties are there in accessing or utilizing these data sources today?***

*While Blue Button 2.0 provides comprehensive access to Medicare Fee-for-Service claims data, several CMS-controlled data sources would significantly enhance the value of the API for developers creating digital products for patients and caretakers.*

***Prior Authorization Data - Submissions, Decisions, and Communications:*** *Beneficiaries cannot see pending authorization requests, approval/denial decisions, or the clinical documentation and communications exchanged These data would help patients understand coverage decisions, track authorization status, and effectively appeal denials. For developers, access to prior authorization data would enable the creation of tools that provide real-time updates and notifications, improving transparency and patient engagement.*

***Cross-reference Data for Care Coordination:*** *A comprehensive list of organizations where beneficiaries have received care, along with organization and provider identifiers, would facilitate direct SMART on FHIR connections. This would help apps discover and connect to clinical data sources, enhancing care coordination. Developers could utilize this data to build applications that*



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*integrate patient records from multiple providers, confirming a more complete and accurate health history.*

***b. What suggestions do you have to improve the Blue Button 2.0 API experience?***

*To enhance the Blue Button 2.0 API experience, several improvements can be made:*

- ***Event-driven Architecture:*** *Implementing an event-driven architecture would significantly improve the API's responsiveness and usability. By adding FHIR Subscriptions, applications could receive real-time notifications when new claims are processed. This would allow patients and caregivers to stay updated on their claims status without having to manually check for updates.*
- ***Alerts for Authorization Decisions and Coverage Changes:*** *Enabling alerts for authorization decisions or coverage changes would provide timely information to patients and their caregivers. This could help them make informed decisions about their care and avoid unexpected coverage issues. For developers, this means creating applications that can proactively inform users about important changes in their healthcare coverage.*

***c. Is there non-CMS data that should be included in the API?***

*While Blue Button 2.0 should remain focused on CMS-controlled data, there are several ways it could facilitate connections to non-CMS data sources to enhance the value of the API for patients and their authorized applications:*

- ***Providing Metadata About Clinical Data Availability:*** *Including metadata about where clinical data might be available, without CMS aggregating that data, would help patients and their authorized applications identify potential sources of valuable health information. This approach utilizes the distributed nature of health data while facilitating connections to complete records.*

*This approach builds on the distributed nature of health data while helping beneficiaries and their authorized applications discover where to connect for complete records.*

***PC-10. How is the Trusted Exchange Framework and Common Agreement (TEFCA) currently helping to advance patient access to health information in the real world?*** *TEFCA's impact on individual patient-initiated access is still nascent. Its potential requires significant evolution towards patient empowerment.*

***Changes Suggested for TEFCA (PC-10b):*** *Our recommendations to advance patient access to health information are:*

- *Empower Individuals with Transparency and Control Over TEFCA Data Sharing: Provide API-accessible audit logs and TEFCA-level patient controls (opt-out, "ask me first," freeze access).*
- *Establish a "TEFCA Patient-Developer Credential" for Comprehensive, Direct Data Access: Offer a cost-free pathway for individuals to use/develop tools for their own data via QHIN APIs.*



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- *Mandate a Trustworthy and Accountable Architecture for All TEFCA Individual Access Services (IAS): Confirm all IAS rely on high-assurance identity verification and explicit, verifiable individual consent, supporting patient-controlled storage models.*
- *EHI as a TEFCA Data Source: Evolve TEFCA to support exchange of full EHI.*

***Impactful Use Cases if Implemented Through a Reformed TEFCA (PC-10c):*** *Patient-initiated aggregation of complete health records; secure sharing with new specialists; individual research with consented data.*

***Adequate Alternatives Outside TEFCA (PC-10g):*** *Direct patient access via certified EHR FHIR APIs.*

***PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSPs)):*** *Wider use of high-assurance digital identity is key to simplifying and securing patient access. However, identity credentials alone are insufficient without robust, bound authorization credentials that specify what data an identified user is permitted to access and for what purpose.*

***Benefits (PC-14b) and Improved Access (PC-14d):***

- *Reduced login fatigue, enhanced security, and critically,* ***simplified and secure account provisioning****.*
- *Use IAL2 CSPs for secure online patient portal account creation.*
- *A strong, federated identity also underpins a "TEFCA Patient-Developer Credential" for Comprehensive, Direct Data Access.*
- *The TEFCA Common Agreement and QTF could mandate a high-assurance security and authorization architecture for all Individual Access Services (whether commercial IAS providers or services facilitating the Patient-Developer Credential). This architecture would confirm that applications accessing data on behalf of an individual do so based on explicit, verifiable individual consent, mediated by a narrow set of trusted identity and authorization service providers, with verifiable binding between identity and authorization.*

***Role in Driving Adoption (PC-14e):***

- ***CMS/ONC:*** *Mandate CEHRT support for IAL2 CSPs for portal account creation/login and encourage for CMS/TEFCA services.*
- ***Providers & Payers:*** *Offer IAL2 CSP-based login options.*
- ***App Developers:*** *Integrate with IAL2 CSPs. Focusing on account provisioning using trusted digital identities is crucial for adoption.*

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***PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?***



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***Provide Hands-On Training and Support.*** *Providers need knowledge and support to effectively use digital health products. CMS/HHS can expand programs to teach staff how to adopt these tools, creating networks with academic centers or quality organizations for on-site and virtual coaching in rural practices. This includes setting up patient portals and using remote monitoring tools. Virtual bootcamps, peer learning collaboratives, and train-the-trainer programs are effective strategies. Public-private collaboration should focus on workflow redesign, data integration, and patient engagement. Additionally, CMS/HHS could create an AI-powered helpdesk for setup and troubleshooting of digital health products.*

***PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)?***
***Please address the following: a. Current challenges in accessing different data formats. b. Impact on patient care quality. c. Technical barriers to full data accessibility.***

- *The core solution is to mandate API-accessible, computable full EHI export, including all EHI (structured, notes, scans, etc.) via API with documentation for computability, overcoming current format-based access barriers.*
- *It is critically important for all data in an EHR to be accessible for exchange to confirm patient safety and effective care.*
- *Missing data negatively impacts patient safety, care coordination, diagnostic accuracy, and efficiency.*
- *Technical barriers are less about format and more about lack of certified capabilities to package and expose all data via APIs, which this recommendation addresses.*

***PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?***

- ***Offer grants for FHIR implementation:*** *Financial support can be a powerful enabler of interoperability. CMS can offer targeted grants or incentive programs to assist providers—especially small practices and rural health systems—in implementing Fast Healthcare Interoperability Resources (FHIR) standards. These grants could cover costs related to upgrading health IT infrastructure, training staff, and integrating FHIR APIs into existing systems. By lowering the financial barrier to entry, CMS can accelerate the*


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*adoption of standardized data exchange protocols and confirm a more equitable rollout of interoperability capabilities across diverse provider settings.*

- ***Align interoperability with existing quality reporting workflows:*** *One of the most effective ways to reduce provider burden is to integrate interoperability requirements into workflows that providers are already familiar with and motivated to optimize—such as quality reporting. This alignment confirms that data sharing becomes a byproduct of routine clinical documentation and performance tracking, rather than an additional administrative task. It also allows providers to use existing EHR capabilities and reporting tools, minimizing disruption and maximizing compliance.*
- ***Use AI to automate data extraction and validation:*** *AI and machine learning technologies can significantly streamline the process of making healthcare data interoperable. CMS can support the development and deployment of AI tools that automatically extract structured data from unstructured clinical notes, validate data accuracy, and map it to standardized terminologies. These tools can reduce manual data entry, improve data quality, and confirm that the information shared across systems is both timely and clinically meaningful. By integrating AI into EHRs and health information exchanges, providers can focus more on patient care and less on data wrangling.*

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***PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?*** *We believe that payers will find TEFCA more valuable if it evolves to prioritize individual control, transparency, and broader innovation. The following limitations hinder payer participation; a member-trusted TEFCA is more valuable to payers.*

***Policy/Technical Limitations and Suggested Changes (Consistent with PC-10):***

- ***Insufficient Individual Control/Transparency:*** *Addressed by empowering individuals with transparency and control over TEFCA data sharing to build member trust.*
- ***Barriers to Innovation for Member-Facing Tools:*** *Lowered by pathways like establish a "TEFCA Patient-Developer Credential" for comprehensive, direct data access.*
- ***Need for Robust Identity/Authorization:*** *Mandate a trustworthy and accountable architecture for all TEFCA Individual Access Services (IAS).*
- ***Limited Data Scope:*** *Expand beyond USCDI by enabling exchange of full EHI.*
- ***Lack of Public Foundational Infrastructure:*** *ONC should lead or actively support the establishment, maintenance, and governance of publicly available, free, and machine-readable national directory services crucial for enabling nationwide health information exchange and interoperability. Effective, scalable interoperability across a diverse national landscape requires common, trusted, and easily accessible infrastructure for discovering*



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*participants, their capabilities, and their electronic endpoints. This reduces friction for all stakeholders, from application developers to HINs and individual patients seeking to connect.*

***PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?*** *We believe a nationwide provider directory with FHIR endpoints that included provider credentialing information (e.g. signed digital assertions about the states in which a provider is licensed to practice) would be immensely valuable to payers.*
***Value to payers:*** *streamlined provider data management, facilitated interoperability for API-based workflows (prior auth, quality data), support for VBC, improved member experience (knowing provider digital capabilities), and enhanced network management.*

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***TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?***
*Developer interest hinges on an open, accessible, and reliable data ecosystem.*
***Short-Term Steps (Next 2 Years):***

- *Aggressively advance mandating API-accessible, computable full EHI export, aligning with industry specifications.*
- *Strengthen single-patient FHIR APIs.*
- *Launch a pilot to establish a "TEFCA Patient-Developer Credential" for comprehensive, direct data access, to lower barriers for individual/small developers.*
- *Commit to and begin developing a public foundational infrastructure for nationwide discovery.*
- *Confirm TEFCA prioritizes individuals by empowering individuals with transparency and control over TEFCA data sharing and mandate a trustworthy and accountable architecture for all TEFCA Individual Access Services (IAS).*
- *To make sure that regulated Bulk FHIR API is a viable and primary mechanism for population data export, rather than a secondary, underperforming option, Certified Health IT implementing FHIR Bulk Data APIs should be foundationally designed to operate efficiently at the population level. In addition, their performance in exporting USCDI data must achieve parity with any proprietary bulk export mechanisms offered by the same system.*

***Longer-Term Steps:***

- *Expand certified API capabilities (e.g., mandate FHIR subscriptions for event-driven workflows and mandate CDS hooks for seamless clinical decision support integration).*


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- *Make access to comprehensive data less about gatekeepers and more about open, standardized interfaces.*

***TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?***

*A nationwide, free, publicly accessible directory of provider FHIR endpoints is foundational. To foster a competitive and innovative health IT market, foundational infrastructure and access pathways must be established on fair and open terms. This includes:*

- ***Publicly Accessible Directories:*** *Core directory services (e.g., for participant endpoints and capabilities) must be publicly available and free of charge to facilitate discovery and interconnection.*
- ***Non-Discriminatory, Cost-Based Access for Commercial Entities:*** *Fees for network participation and data access services for commercial entities should be reasonable, non-discriminatory, and based on the actual costs of providing those services, preventing anti-competitive pricing.*
- ***Cost-Free Pathways for Individual Innovation:*** *Clear, secure, and cost-free pathways must exist for individuals to access their own data using tools they choose or develop themselves, confirming that personal innovation and patient-driven solutions are not stifled by commercial fee structures.*

*ONC should lead/govern. The directory must be publicly available and free of charge to maximize utility and adoption.*

***TD-6. What unique interoperability functions does TEFCA perform?***
***a. What existing alternatives should be considered? b. Are there redundant standards, protocols or channels or both that should be consolidated?***

*TEFCA's intended unique function is nationwide querying of unknown data holders under a common trust agreement. Its current realization needs strengthening.*

***To bolster unique value:***

- *Empower individuals with transparency and control over TEFCA data sharing*
- *Establish a "TEFCA Patient-Developer Credential" for comprehensive, direct data access*
- *Mandate a trustworthy and accountable architecture for all TEFCA Individual Access Services (IAS)*

***Existing Alternatives (TD-6a):***

- *Direct EHR FHIR APIs; regional/state HIEs; proprietary vendor networks.*
- *TEFCA should complement, not replace, direct patient-to-EHR API access, offering value for discovery, provided it fully embraces patient empowerment.*



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***TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?***
***a. Does it contain the full extent of data elements you need? b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized? c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed? d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?***

*USCDI is a valuable baseline but limited in scope and granularity. ONC must lead an improved, evidence-based USCDI development and adoption process to confirm that an expanding set of meaningfully standardized and clinically relevant patient data elements is defined. This enhanced USCDI will serve as the common data foundation for all mandated FHIR-based APIs.*

1. ***Evidence-Based Prioritization:*** *The process for expanding USCDI must prioritize data elements with demonstrated, widespread real-world use cases and significant benefits for patient care, interoperability, research, or public health.*
2. ***Clear Functional Expectations for SDOs:*** *When new data elements are added to USCDI, ONC should clearly articulate the functional expectations and provide illustrative examples of intended real-world usage. This high-level guidance enables ONC to work effectively with Standard Development Organizations (SDOs) to drive the downstream development of detailed, API-specific technical specifications and implementation guides (e.g., US Core FHIR profiles for single-patient access, and relevant profiles for Bulk FHIR operations).*
3. ***Iterative Refinement:*** *An active feedback loop post-USCDI version release should be established to assess implementation quality and consistency across all API types, allowing for refinement of functional expectations or guidance.*
4. ***Conformance Testing:*** *Certification testing for all FHIR APIs mandated to expose USCDI data must verify that the implementation of USCDI data elements aligns with the detailed specifications in relevant SDO-developed FHIR Implementation Guides (e.g., US Core), serving as a practical proxy for confirming consistency with ONC's articulated functional expectations.*

***Limitations (TD-7a, TD-7b):***

*Primarily scope; USCDI is intentionally a "core" set.*

***Adding More Data Elements to USCDI (TD-7c):***

*Yes, thoughtfully adding more elements via the process described above adds value. Address scoping via iterative expansion and clear value propositions.*

***Less Structured Formats and LLMs (TD-7d):***

*We need **both**: expanding standardized USCDI and API access to complete EHI (including less structured data). LLMs can process the unstructured parts of EHI, while standardized USCDI remains vital for precision tasks.*



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***TD-9. Regarding certification of health IT:***

***a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?***

*Prioritizing API capabilities over internal functionality would:*

- *Enable innovation at the application layer*
- *Focus market competition on improved user experience*
- *Confirm consistent data access regardless of EHR choice*
- *Simplify certification to measurable outcomes*

***c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?***

*Certification must require full EHI Export APIs and Programmatic access to Imaging Data:*

- *API access to all stored electronic health information, with documentation for any proprietary formats*
- *Programmatic image access through standardized protocols (may require criteria specific to PACS as a subset of Certified Health IT).*
- *Performance requirements for population-scale use*

***e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?***

*EHRs with robust bulk data transfer could dramatically reduce provider burden in CMS quality reporting by moving away from current flawed methods. Today's digital quality measures often assume EHRs provide data with perfect consistency. However, even for USCDI-defined elements like "clinical assessments," uniform population across all systems and settings is not guaranteed. This mismatch between measure assumptions and real-world data accessibility means care might be delivered but not credited due to rigid technical specifications. This leads to manual abstraction, separate quality management portals, and extra "checklist" tasks for clinicians.*

*A better approach involves:*

- *More Flexible Quality Measures: Design measures focused on clinical intent rather than narrow data fields.*
- *Advanced Analytics on Comprehensive Data: Employ an intelligent analytical layer (e.g., LLM-powered) to assess quality by analyzing:*
    - *Structured USCDI data via Bulk API;*
    - *Unstructured USCDI data via Bulk API;*
    - *Clinical notes via Document References, interpreted by LLMs;*



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- *Full Electronic Health Information (EHI) Export to capture data beyond standard USCDI profiles, filling critical gaps.*

*This shifts the burden from manual, point-of-care compliance to automated, post-hoc analysis of bulk-exported data, allowing natural clinical documentation. Essential for this shift is confirming bulk data API capabilities and performance.*

***TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?***

*The Cures Act's vision of data being accessed, exchanged, and used "without special effort" extends beyond simple retrieval. It encompasses the full lifecycle of patient interaction with their data, including confirming its accuracy and completeness.*

*Mandating an API-accessible, computable full EHI export, aligning with industry specifications, will fulfill the Cures Act's "all data elements... without special effort" provision by requiring API accessibility, inclusion of all EHI, and computability via vendor documentation.*

***Further Supporting Recommendations confirming "without special effort" for access and use:***

- *Keep Single-Patient API Certification Current with SMART App Launch & Backend Services Specifications: The ONC Health IT Certification Program must confirm certified Health IT systems implement current, stable, industry-adopted versions of the SMART App Launch Framework and related backend services specifications to support secure and functional single-patient FHIR API access. Confirms modern, secure, and functional single-patient API access.*
- *Mandate API Access to Full EHI. Patients require API access to all their electronic health information (EHI)—including notes and images—in a computable format. This empowers patients and aligns with industry standards like the Argonaut EHI Export API for innovative applications.*
- *Confirm Programmatic and Automated Access to Medical Images: Certified Health IT must provide programmatic and automatable API access to diagnostic quality medical images, using consistent, standardized authorization flows and confirming images are shareable and usable by authorized applications. Makes critical image data accessible without special effort.*
- *Mandate Electronic Pathways for Patient Record Amendment Requests: Certified Health IT must provide clear, secure, and entirely electronic pathways for individuals to request amendments to their medical records, track the status of these requests, and receive*



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*responses, thereby making HIPAA-granted rights more accessible and usable. These pathways must be available through patient-facing interfaces (e.g., patient portals) and programmatically via APIs for patient-authorized applications. Fulfilling the HIPAA right to request amendment "without special effort" is a crucial aspect of "using" one's health information. Current manual processes create significant burdens. Mandating certified electronic pathways (via portals and APIs) for patients to submit, track, and receive responses to amendment requests directly aligns with the Cures Act's intent to empower patients and improve data quality.*

- *Mandate Patient-Initiated Secure Messaging via Standardized APIs: Certified Health IT must support secure, patient-initiated messaging to healthcare providers from third-party applications, utilizing standardized APIs. Signal intent to adopt community-developed standards such as the Argonaut Project's Provider/Patient Secure Messaging API Implementation Guide (https://hackmd.io/@argonaut/H1dQ95xG3) via SVAP. Enabling patients to communicate with providers directly from their chosen apps, potentially with relevant data context, greatly increases the utility and stickiness of digital health products.*

*By mandating these capabilities through the ONC Health IT Certification Program, ONC can confirm that "without special effort" becomes a practical reality for patients seeking to truly engage with and manage their complete health information.*

***TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10))...***
***a. Should this capability be revised to specify standardized API requirements for EHI export? b. Are there specific workflow aspects that could be improved? c. Should CMS consider policy changes to support this capability's use?***
*Yes, the EHI export capability urgently needs revision to specify standardized API requirements.*
***Standardized API Requirements for EHI Export (TD-11a):***
***Yes, unequivocally.*** *The current non-API approach is insufficient. ONC should require alignment with or equivalence to the* ***Argonaut Project's EHI Export API IG****, as detailed in our recommendation.*
***Workflow Aspects for Improvement (TD-11b):***

- *Patient-initiated API-driven workflow*
- *Electronic request initiation*
- *Clear API-based status tracking*

***CMS Policy Changes to Support Use (TD-11c):***
*Promote beneficiary awareness, confirm TEFCA can eventually support full EHI exchange, and reinforce that API-accessible EHI export must be free to patients.*


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***VB-3. What are essential health IT capabilities for value-based care arrangements? a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools. b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?***

*VBC success depends on timely, comprehensive data access, robust analytics, and proactive engagement.*

***Essential Health IT Capabilities Supported by Our Recommendations:***

- *Efficient Data Extraction/Aggregation*
- *Timely Patient Event Notifications*
- *Advanced CDS/Workflow Integration*
- *Comprehensive Data for Quality Measurement and Bulk FHIR capabilities.*
- *Enhanced Patient Engagement and Data Access*
- *Nationwide Data Discovery and a reformed TEFCA*

***VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?***

*A nationwide provider directory of FHIR endpoints would greatly benefit VBC organizations.*

***Benefits for VBC Organizations:***

- *Improved data access for attributed populations, facilitated care coordination, better understanding of claims data by linking to clinical sources, support for transitions of care, and identification of technically capable partners.*

***Key Data Elements:***

- *FHIR API base URLs, supported FHIR versions/IGs, TEFCA participation details, authentication mechanisms, organizational affiliations, and certified Health IT product info.*



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